

Control of propellant slosh dynamics in observation spacecraft using Model Free Control and pressure sensors

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Abstract—The mitigation of fuel propellant sloshing on-board satellites or during space launches is of particular interest. Fuel sloshing can indeed significantly affect spacecraft motion by increasing the maneuver settling time and occasionally destabilizing them. This paper presents a novel, model-free attitude control approach that actively compensates for the sloshing disturbance. One of the main novelties consists of using pressure sensors inside the fuel tank to measure a quantity proportional to the sloshing torque. Then, using this measurement, an attitude controller can be designed not only to reject the sloshing torque but also to keep the fuel sloshing within acceptable conditions. Technically speaking, this is achieved by combining model-free control with an output to input saturation transformation (OIST) add-on.

I. INTRODUCTION

Many spacecraft and space launchers rely on large propellant tanks to perform attitude control, orbit insertion, and station-keeping maneuvers. Since a large percentage of the spacecraft mass is composed of fluid, liquid propellant sloshing can significantly affect spacecraft dynamics and stability. The complex fluid dynamics in micro gravity have caused mission failures in the past [1], [2] and can limit spacecraft performance and agility. Therefore, mitigating sloshing effects through proper control algorithms is an important aspect of spacecraft design, which often requires slowing down the spacecraft and assuming conservative margins when performing maneuvers.

Classical methods have modeled sloshing fluid using an Equivalent Mechanical Model [3], [4], [5], [6] and use this approximation to compensate for the non-linear perturbations of sloshing in a fuel tank. More advanced models have used direct computational fluid dynamic (CFD) models in the fuel tank itself to predict the sloshing effects [7]. Still, they are generally too slow to be used in closed-loop simulations. That is why new control oriented models were recently proposed in [8]. In particular, these models were used to synthesize new robust linear parameter varying (LPV) observers [9], [10] to both estimate and compensate for the sloshing torque. Although the models were uncertain, their structure was still known.

In this paper, the problem of actively compensating for sloshing is addressed when the control engineer does not have any a priori knowledge on the structure and dynamics of the fuel sloshing. First, we propose to compensate for the fluid disturbance by using the Model Free Control approach of [11]. Second, we propose to use pressure sensors placed

inside the fuel tank for use in an add-on controller, which limits the agitation of the fluid. An example of pressure sensors placed inside the fuel tank is illustrated in Figure 1.

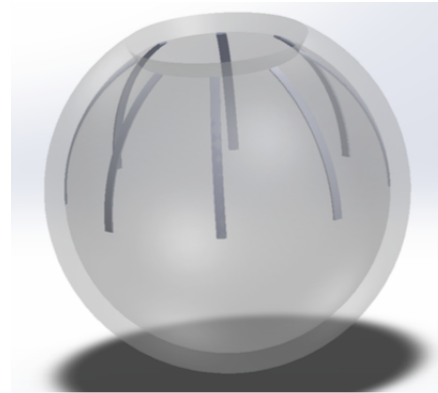


Fig. 1. Strip style pressure sensors inside tank

The add-on controller is based on an output to input saturation transformation [12], which converts the output constraint into control input time-varying bounds. Such a transformation was already used in the past to limit the angular velocity of a spacecraft equipped with flexible panels [13]. The design was quite complicated when the disturbance torque was not measured. In this paper, we propose to combine the model free control method with such a transformation to facilitate its implementation when the model is poorly known and the perturbation is not measured.

The paper is organized as follows: first, we provide a few preliminaries on the model free control and output to input saturation transformation methods. Then, in Section III, we present two different dynamical models that represent a spacecraft disturbed by liquid fuel. A generic control strategy is proposed in Section IV to mitigate the fuel sloshing when either model is used. Finally, simulation results illustrate the efficacy of the proposed control method.

II. PRELIMINARIES

A. Model Free *i*-PD Control

Let us here summarize the main ideas of the so-called intelligent proportional derivative controller (*i*-PD) introduced in [11] and since then successfully applied to many domains (see e.g [14], [15], [16]).

Consider an unknown finite dimensional system and suppose that the following input/output relation exists between a

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single control input (u) and single output (y) in a differential equation formulation:

$$\mathbb{E}(y, \dot{y}, \dots, y^{(a)}, u, \dots, u^{(b)}) = 0 \quad (1)$$

where $\mathbb{E}$ is a polynomial function with real unknown coefficients. Assuming we do not have any information about the plant, we follow the ideas of [11] and approximate this relation by a second-order *Ultra-Local* model of the form:

$$\ddot{y}_m = F_y + \lambda u \quad (2)$$

where y_m is the measured output and F_y is piecewise constant. Using the collected measurement of y_m and the applied control input u in the last T seconds, it can be shown [11] that F_y can be approximated by

$$\begin{aligned} \hat{F}_y(t) = & \frac{5!}{2T^5} \int_{t-T}^t \{ [T^2 - 6T(t-\sigma) + 6(t-\sigma)^2] y_m(\sigma) \\ & - \frac{\lambda}{2} [T - (t-\sigma)]^2 (t-\sigma)^2 u(\sigma) \} d\sigma \end{aligned} \quad (3)$$

Using this approximation, one can directly reject this matched disturbance using the control input $u = -\frac{1}{\lambda} \hat{F}_y$ and also add to it a feedforward term $\frac{1}{\lambda} \ddot{y}_d$ and a PD (or PID) feedback term to make y_m track the desired trajectory y_d . These controllers are usually called intelligent PD or PID controllers and noted as i-PD or i-PID.

B. Model Free Output to Input Saturation Transformation

The output to input transformation was recently proposed to convert an output constraint into an input saturation [12]. Similarly to other constrained control methods like for example reference governors [17], this method cannot easily be used when the system dynamics is uncertain or when its state is not fully measured [13], [18]. Though, this transformation can be easily applied when an *Ultra-Local* model is used to approximate the unknown dynamics during a 'short time' window. Let us again consider the following second order *Ultra-Local* model as defined in [11]:

$$\ddot{y}_m = F_y + \lambda u \quad (4)$$

where $\lambda \neq 0$ is constant.

Let $y_{\min} < y_{\max}$ be the constant limitations one would like to impose to the output y_m .

It is easy to see that if $\forall t \geq 0, \dot{y}_m \in [-\kappa_1(y_m - y_{\min}), -\kappa_1(y_m - y_{\max})]$ with $\kappa_1 > 0$ and $y_m(0) \in [y_{\min}, y_{\max}]$ then $\forall t > 0, y_m(t) \in [y_{\min}, y_{\max}]$. Then, the same idea can be used to limit $\dot{y}_m$ between $y_{\min} < \dot{y}_{\max}$ where

$$\dot{y}_{\min} = -\kappa_1(y_m - y_{\min}) \quad \& \quad \dot{y}_{\max} = -\kappa_1(y_m - y_{\max}) \quad (5)$$

Let us now note $\ddot{y}_{\min}$ (resp. $\ddot{y}_{\max}$) the time derivative of $\dot{y}_{\min}$ (resp. $\dot{y}_{\max}$). Again, one can easily see that if $-\kappa_2(\dot{y}_m - \dot{y}_{\min}) \leq \ddot{y}_m - \ddot{y}_{\min}$ and $\ddot{y}_m - \ddot{y}_{\max} \leq -\kappa_2(\dot{y}_m - \dot{y}_{\max})$ with $\kappa_2 > 0$ then $\dot{y}_m(t)$ will remain inside $[\dot{y}_{\min}, \dot{y}_{\max}]$ provided $\dot{y}_m(t)$ lies in this interval at the initial instant. The obtained bounds on $\ddot{y}_m = F_y + \lambda u$ can thus be applied to the control input u when λ is known and F_y is approximated by $\hat{F}_y$ (the approximation is so accurate that we neglect the approximation error [11]).

III. MODELING

This section presents two different models on which one can apply the same and new model-free control method described in the sequel. Ultimately, we'd like to apply this control method to a CubeSat equipped with a water tank without using a dynamical model to tune the controller.

A. Equivalent Mechanical Model

This subsection formulates the dynamics of a partially filled fuel container with a reaction wheel including the prominent liquid slosh modes. It is inspired by [5]. The fuel container is represented as a rigid body. The fuel sloshing is modeled using a multi-mass-spring-damper system, where the natural frequencies of the mass-spring elements represent the prominent sloshing modes.

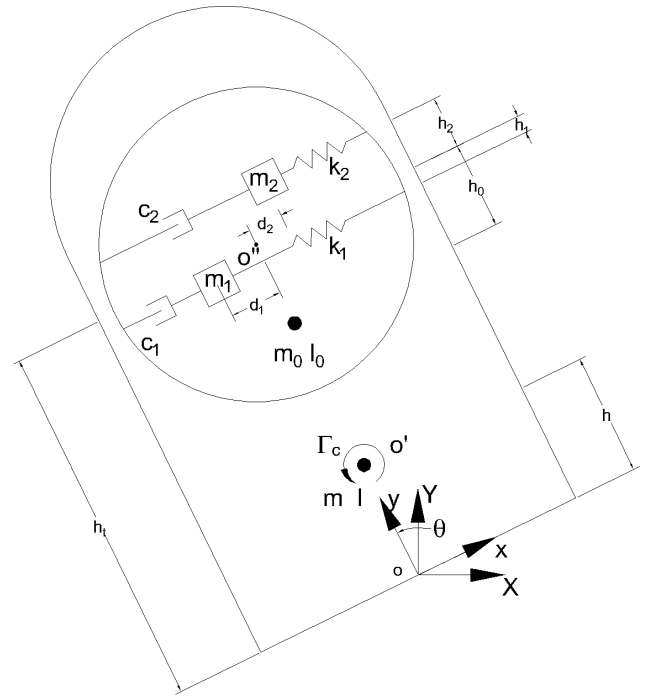


Fig. 2. A multiple mass-spring slosh model for a spacecraft

We introduce an inertial reference frame, XoY , of which its origin o is located at the center of the bottom of the spacecraft and a body-fixed reference frame, xoy , with the same origin o as indicated in Figure 2. The angle θ denotes the rotation from the inertial reference frame to the body-fixed reference frame. The point o' , a distance h over the point o along the body-fixed y -axis, denotes the location of the center of mass of the spacecraft and the fuel tank together, whose mass and moment of inertia (with respect to its own center of mass) are m and I , respectively. The fuel is modeled by a rigid body m_0 and point masses m_i , $i = 1, \dots, N$, whose relative position along the body-fixed x -axis are denoted by d_i . The longitudinal locations with respect to the center of the tank are denoted as h_0 and h_i , respectively. The moment of inertia assigned to the rigid body is I_0 , and to the point masses are usually negligible. The

spring constant $k_i, i = 1, \dots, N$ is determined with respect to the slosh natural frequency of the i th mode. In general, k_i would be a function of the axial acceleration. However, in this paper, it is assumed that the system does not operate aggressive maneuver so that k_i remains constant. A torque Γ_r acting on the center of mass of the spacecraft is the control input. The center of the tank is located a distance h_t over the origin o along the body-fixed y -axis. To preserve the static properties of the fuel, the sum of all the masses must be the same as the fuel mass m_f , and the center of mass of the model must be at the same elevation as the fuel, i.e.

$$m_0 + \sum_{i=1}^N m_i = m_f, \quad (6)$$

$$m_0 h_0 + \sum_{i=1}^N m_i h_i = 0. \quad (7)$$

Let $\hat{i}$ and $\hat{j}$ be the unit vectors along the body-fixed transverse and longitudinal axes, respectively. The position vector of the center of mass of the spacecraft and the fuel tank can be expressed as,

$$\vec{r}_{o'} = h\hat{j}. \quad (8)$$

Clearly, the velocity vector can be computed as,

$$\dot{\vec{r}}_{o'} = -h\dot{\theta}\hat{i}. \quad (9)$$

Similarly, the position vectors of the fuel masses $m_0, m_i, i = 1, \dots, N$ can be expressed respectively as,

$$\vec{r}_0 = (h_t + h_0)\hat{j}, \quad (10)$$

$$\vec{r}_i = d_i\hat{i} + (h_t + h_i)\hat{j}, \quad \forall i. \quad (11)$$

Assuming h_i are constants, the velocity vectors can be computed as,

$$\dot{\vec{r}}_0 = -(h_t + h_0)\dot{\theta}\hat{i}, \quad (12)$$

$$\dot{\vec{r}}_i = [\dot{d}_i - (h_t + h_i)\dot{\theta}]\hat{i} + d_i\dot{\theta}\hat{j}, \quad \forall i \quad (13)$$

The total kinetic energy can be expressed as,

$$T = \frac{1}{2}m\dot{\vec{r}}_{o'}^2 + \frac{1}{2}m_0\dot{\vec{r}}_0^2 + \frac{1}{2}(I + I_0)\dot{\theta}^2 + \frac{1}{2}\sum_{i=1}^N(m_i\dot{\vec{r}}_i^2). \quad (14)$$

The total potential energies of the spacecraft, the fuel tank and the fuel can be expressed as,

$$U = mgh \cos \theta + \sum_{i=0}^N m_i g(h_t + h_i) \cos \theta + \frac{1}{2} \sum_{i=1}^N k_i d_i^2. \quad (15)$$

Under the indicated assumptions, the Lagrangian is expressed as,

$$L = T - U. \quad (16)$$

The damping constants $c_i, i = 1, \dots, N$ are introduced to model the dissipative effects due to the fuel slosh. A fraction of the kinetic energy of a sloshing fuel is dissipated during each cycle of the motion. In Lagrangian mechanics,

the Rayleigh dissipation function R is exploited to describe the dissipative effects, which can be expressed as,

$$R = \frac{1}{2} \sum_{i=1}^N c_i \dot{d}_i^2. \quad (17)$$

Applying Lagrange's formulation and the first two modes of fuel sloshing,

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = \Gamma_c, \quad (18)$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{d}_i} - \frac{\partial L}{\partial d_i} + \frac{\partial R}{\partial \dot{d}_i} = 0, \quad i = 1, 2, \quad (19)$$

where Γ_c is the control input torque.

The equations of motion can be obtained as,

$$\begin{aligned} & \sum_{i=1}^2 m_i \{ [(h_t + h_i)^2 + d_i^2] \ddot{\theta} + 2d_i \dot{d}_i \dot{\theta} - (h_t + h_i) \ddot{d}_i \} \\ & + [mh^2 + m_0(h_t + h_0)^2 + I + I_0] \ddot{\theta} \\ & - [mh + \sum_{i=0}^2 m_i(h_t + h_i)] g \sin \theta = \Gamma_c, \end{aligned} \quad (20)$$

$$m_i \ddot{d}_i + c_i \dot{d}_i + k_i d_i - m_i(h_t + h_i) \ddot{\theta} - m_i d_i \dot{\theta}^2 = 0, \quad i = 1, 2. \quad (21)$$

Since the positions d_i are very difficult or even impossible to measure in reality, we take into consideration the pressure of the fuel acting on the tank to be one type of measurement to control the system. To simplify the modelling, because of the symmetry of the structure, let P_l and P_r denote the mean pressure of the left-hand and right-hand side of the fuel acting on the tank along the transverse direction in the body-fixed reference frame. Applying Newton's second law and D'Alembert's principle, the equation of motion can be obtained as,

$$\sum_{i=1}^2 m_i \ddot{d}_i + c_i \dot{d}_i + k_i d_i = (P_l - P_r)A + \sum_{i=0}^2 m_i(h_t + h_i) \ddot{\theta}, \quad (22)$$

where A is the largest cross-section area of the fuel, in static equilibrium, along the longitudinal axes in the body-fixed reference.

Substituting the equation (21) into the equation (22), the equation can be rewritten as,

$$P_r - P_l = \frac{1}{A} [m_0(h_t + h_0) \ddot{\theta} - \sum_{i=1}^2 m_i d_i \dot{\theta}^2], \quad (23)$$

Taking into account the sensor dynamics, we finally measure:

$$\Delta_P = \frac{5}{s^2 + 2s + 5} (P_r - P_l) \quad (24)$$

B. More general control oriented model

The second model is again a single axis model. This one is based on the recent work presented in [8], [9], [10]

which created time varying models based on the CFD code 'DIVA'[7]. The model is of the following form:

$$J\ddot{\theta} = \Gamma_r + \Gamma_s \quad (25)$$

$$\ddot{\Gamma}_s = -a_S(t)\dot{\theta} - b_S(t)\ddot{\theta} - c_S(t)\Gamma_s - k_S(t)\dot{\Gamma}_s \quad (26)$$

$$\Gamma_r = \frac{1.2s + 0.75}{s^2 + 2.4s + 0.75}\Gamma_c \quad (27)$$

where J and θ denote the inertia of the satellite and its attitude angle respectively. Three different torques apply to the attitude angle dynamics: Γ_r is the control torque realized by the reaction wheel whose dynamics are given by (27), Γ_c is the control torque input, and Γ_s is the fuel sloshing torque. Contrary to the traditional Equivalent Mechanical Models, the bounded time-varying coefficients a_S, b_S, c_S, k_S of this model depend on the angular velocity $\dot{\theta}$ and acceleration $\ddot{\theta}$ of the satellite. It was indeed observed that the dynamics of bubbles sloshing are highly dependent on these two parameters in microgravity [7].

Pressure sensors are placed within the fuel tank as depicted in Figure 1. Similar to what was discussed at the end of the previous section, we assume the pressure sensors measurements can be post-processed to obtain a quantity that represents the agitation of the fluid and which cancels when $\Gamma_s = \dot{\Gamma}_s = \Gamma_r = 0$. We denote it again by Δ_P .

IV. CONTROL STRATEGY

Using second-order Ultra-Local models, the two models described in the previous section can be approximated by a model of the following form:

$$\ddot{\theta} = \lambda_1\Gamma_c + F_1(t) \quad (28)$$

$$\ddot{\Delta}_P = \lambda_2\Gamma_c + F_2(t) \quad (29)$$

The objective of the control synthesis is twofold:

- 1) In the absence of constraints, the controller must nominally compensate the sloshing torque disturbance $F_1(t)$ and make θ follow the desired trajectory $\theta_d(t)$.
- 2) Given $\Delta_{\min} < 0 < \Delta_{\max}$, the controller must keep Δ_P within the interval $[\Delta_{\min}, \Delta_{\max}]$.

A. Nominal i-PD Controller

The goal of the nominal controller is to track the desired continuous acceleration profile $\ddot{\theta}_d$, initially defined according to the mission objectives. Next, θ_d is simply obtained by double integration of $\ddot{\theta}_d$. A standard proportional derivative controller is initially tuned by pole placement on the second-order Ultra-Local model (28) when $F_1(t) = 0$:

$$\Gamma_c = \frac{1}{\lambda_1}\ddot{\theta}_d - K_p(\theta - \theta_d) - K_d(\dot{\theta} - \dot{\theta}_d) \quad (30)$$

Noting $\Gamma_d = \frac{1}{\lambda_1}\ddot{\theta}_d + K_p\theta_d + K_d\dot{\theta}_d$, we rewrite the PD controller as follows:

$$\Gamma_c = \Gamma_d - K_p\theta - K_d\dot{\theta} \quad (31)$$

Using the collected measurement of θ and Γ_c , we calculate at every instant $t \geq T$ an estimate $\hat{F}_1(t)$ of $F_1(t)$ using the

equation (3) and finally obtain the following nominal i-PD (intelligent Proportional Derivative) controller:

$$\Gamma_c^{nom} = \Gamma_d - K_p\theta - K_d\dot{\theta} - \frac{1}{\lambda_1}\hat{F}_1 \quad (32)$$

B. Constrained Controller

Now, we are interested in bounding the pressure sensors difference Δ_P . One chooses the bounds on Δ_P so that the fluid and, thus the satellite, remain within a safe envelope. Handling the Δ_P constraints begins with considering the Ultra-Local model (29) and applying the output to input saturation transformation. As discussed in the preliminaries, Δ_P will remain inside its 'safe' interval provided it lies in this interval at the initial instant, $\dot{\Delta}_P(0) \in [\dot{\Delta}_{\min}(0), \dot{\Delta}_{\max}(0)]$ and that for all $t \geq 0$, $\ddot{y}$ satisfies

$$-\kappa_2(\dot{\Delta}_P - \dot{\Delta}_{\min}) + \ddot{\Delta}_{\min} \leq \ddot{\Delta}_P \leq -\kappa_2(\dot{\Delta}_P - \dot{\Delta}_{\max}) + \ddot{\Delta}_{\max} \quad (33)$$

which can be rewritten as follows:

$$\ddot{\Delta}_P \in -(\kappa_1 + \kappa_2)\dot{\Delta}_P - \kappa_1\kappa_2(\Delta_P - [\Delta_{\min}, \Delta_{\max}]) \quad (34)$$

clearly, when $\Delta_P \in [\Delta_{\min}, \Delta_{\max}]$, we have $\dot{\Delta}_P \in [-\kappa_1(\Delta_{\max} - \Delta_{\min}), -\kappa_1(\Delta_{\min} - \Delta_{\max})]$ and therefore the $\dot{\Delta}_P$ limitations, though state dependent, are bounded. Finally, applying the OIST add-on to the nominal i-PD controller consists in saturation Γ_c such that $\lambda_2\Gamma_c$ is kept within the following interval:

$$\lambda_2\Gamma_c \in -\hat{F}_2(t) - (\kappa_1 + \kappa_2)\dot{\Delta}_P - \kappa_1\kappa_2(\Delta_P - [\Delta_{\min}, \Delta_{\max}]) \quad (35)$$

Let us note $[\Gamma_{c,\min}(t), \Gamma_{c,\max}(t)]$ the interval in which Γ_c must lie. This interval has some time-varying bounds but its width is constant and equal to $\frac{1}{|\lambda_2|}(\Delta_{\max} - \Delta_{\min})$. This means that its bounds cannot cross over.

C. Closed-loop system

Finally, our controller is based on the combination of an i-PD controller with an output to input saturation add-on. It is written as follows:

$$\Gamma_c(t) = \max(\min([\Gamma_c^{nom}(t), \Gamma_{c,\max}(t)], \Gamma_{c,\min}(t))) \quad (36)$$

This controller is applied to the system (28)-(29). Closed-loop stability properties could be established under mild assumptions on $F_1(t)$ and $F_2(t)$ (in particular if one assumes that they remain bounded). The use of an anti-windup loop could also extend the region of attraction and ease the stability proofs (see [12], [13]). However, our approach has some limitations which could prevent using such assumptions on $F_1(t), F_2(t)$. As an example, the bandwidth of the reaction wheel model (27) prevents the controller from using too significant gains $K_p, K_D > 0$ for the nominal controller or too significant gains $\kappa_1, \kappa_2 > 0$ for the OIST add-on. A more in-depth study of the closed-loop system will be the subject of our future work.

V. NUMERICAL RESULTS

To illustrate the efficacy of our approach, the proposed Equivalent Mechanical Model with an i-PD controller and output to input saturation add-on was simulated using the MATLAB programming environment. The parameters chosen for the simulations are tabulated in Table I. To demonstrate our approach, we first applied the nominal controller in order to make θ track a desired sinusoidal signal $\theta_d(t)$, which is chosen as,

$$\theta_d = \frac{10^\circ \times 2\pi}{360^\circ} \sin(0.2\pi t). \quad (37)$$

We observed that $|\Delta_P|$ reached 32 Pa and then chose to limit it within $[-23, 23]$ Pa. This choice is, at present, arbitrary and will be the scope of future studies. This is a design parameter for the control engineer and corresponds to the maximal allowable 'fuel agitation' he/she would like to enforce.

Simulation results are represented in the following figures. In Figure 3, it is observed that the proposed control system can track the desired trajectory within 8 seconds. At the same time, in Figure 4, $|\Delta_P|$ is successfully kept between the desired bounds, which are plotted in red. Twice, from 0 to 2 seconds and from 2 seconds to 4 seconds, $|\Delta_P|$ approaches the bounded constraints but does not violate them. This fact is also illustrated in Figure 5. Indeed, in Figure 5, the control input torque (in blue) is nominally based on the Model Free Control (MFC) method, but when $|\Delta_P|$ approaches its bound, the output to input saturation transformation method is activated so that the actual control input torque overlapped with either the maximum torque bound (orange line) or the minimum torque bound (yellow line).

VI. CONCLUSIONS

The goal of this work was two-fold: to demonstrate an effective method for rejecting the fuel sloshing torque onboard observation satellite and to show that pressure sensors can be used to better mitigate the fuel sloshing disturbance by limiting its amplitude. By way of the two modeling examples, this work illustrates that one can combine model-free control with a constrained control method without using a dynamical model but merely employing 'Ultra-Local' models. Here, we

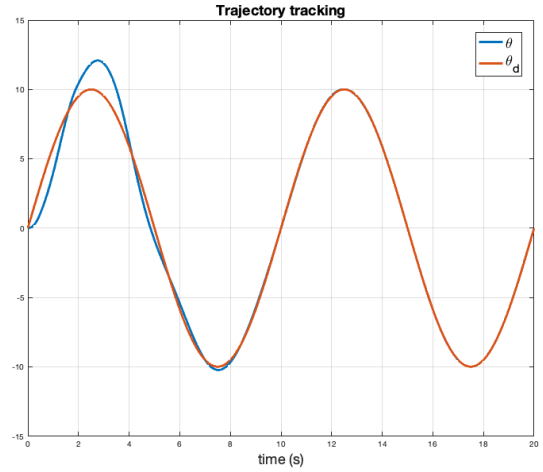


Fig. 3. Time response of θ .

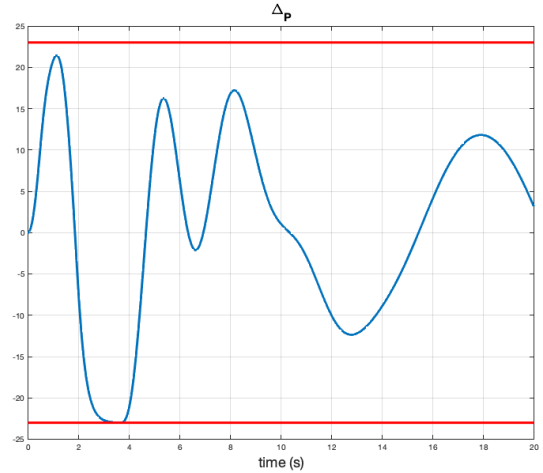


Fig. 4. Time response of the pressure sensor output.

TABLE I
NUMERICAL SIMULATION PARAMETERS

<i>Fuel Sloshing</i>		<i>Spacecraft</i>	
Parameter	Value	Parameter	Value
m_0	205 kg	m	975 kg
m_1	195 kg	I	400 kg·m ²
m_2	6.5 kg	h	1.2 m
I_0	44.1 kg·m ²	h_t	0.6 m
h_0	-0.135 m	<i>Controller</i>	
h_1	-0.145 m	Parameter	Value
h_2	0.035 m	K_p	5
k_1	1174 N/m	K_d	1
k_2	120 N/m	λ_1	5.284×10^{-4} 1/(kg·m ²)
c_1	48 N·s/m	λ_2	0.0866 1/m ³
c_2	2.75 N·s/m	κ_1	5
A	0.5817 m ²	κ_2	5

used the output to input saturation transformation strategy to handle the output constraint. Additionally, other constrained control approaches like MPC (Model Predictive Control) or Reference Governors could be studied in the future. Our method was shown to be effective using a numerical example as it enabled pointing of the satellite toward a new direction while limiting the pressure sensors measurements. The computation of the region of attraction of the closed-loop system will be the subject of future studies.

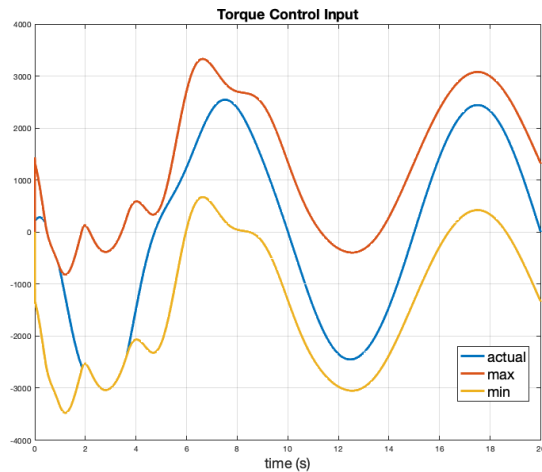


Fig. 5. Time response of the torque control input.

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